Moving the Wall

Coupling detection between two noise-like streams, from baduk to Granger causality

Abstract. Two time-dependent streams can each look like random noise and still be bound tightly together. Baduk lets me prove it on a case with ground truth: one color's sequence of coordinates is nearly unpredictable on its own, yet it answers the other color move by move, so the whole game is a single coupled object whose truth I can always check. I drafted an interpretable and compact predictor, but it plateaued, so I stopped trying to predict and measured the coupling directly. The generalized meter is exactly Granger causality, and by a theorem of Barnett, Barrett, and Seth (2009) Granger is the wall of linear coupling: for jointly Gaussian processes it equals transfer entropy, with nothing recoverable past it. Baduk, however, is not Gaussian, and a board-aware lens recovers a small, recognizable coupling the linear meter cannot see, the local response of white answering black at close range. A richer lens recovers a little more of the same coupling, and a little more again, the leftover always smaller but never gone. The same decomposition should travel to any system whose agents respond to one another's positions, from animal movement to, in principle, team sports.

Terminology. The game studied in this paper is 바둑 / Baduk (Korean), 圍棋 / Wéiqí (Chinese), 囲碁 / Igo (Japanese, rendered as Go in Western literature). The corpus is drawn predominantly from Korean and East Asian professional play; I follow the Korean convention and use Baduk throughout.

1. Introduction

In baduk, two players alternate stones on a 19×19 grid. If I discard the rules and follow only one color's coordinates over time, the sequence is nearly unpredictable from its own past; to a second-order linear lens it looks like noise. The same holds for the other color. Yet the two colors are plainly not independent. White answers black and black presses white, so the game is a single coupled object. This is the situation I set out to study: two individually noise-like streams that are jointly determined. Baduk supplies it with an answer key, because the true course of play is always available to check against.

The motivating measurement is simple. Across thousands of professional games, black's most recent move explains roughly 44% of the predictable part of where white plays next, beyond what white's own history already supplies. Apart, the streams resemble noise. Together, they are coupled. The questions this paper pursues are how to quantify that coupling honestly, what fundamental limit governs it, and how far a real system lives past that limit.

The coupling I measure is the relation Granger (1969) formalized and Schreiber (2000) recast in information terms as transfer entropy. My contribution is not a new estimator. It is a re-derivation of that machinery from an unusual starting point, a verification that it behaves exactly as theory requires, and a controlled measurement of how far baduk's coupling extends past the linear-Gaussian limit.

2. Methods

2.1 From a predictor to a coupling meter

My first approach was to predict the next move with a compact, interpretable model, here called E1: a conditional logit over the 361 cells. The idea is to score every empty cell z as a candidate for the next move and turn those scores into a probability distribution. The score, or utility, of a cell is a per-cell bias plus a weighted sum of local-geometry features, and the predicted move is a softmax over the utilities,

$$U(z) = b_z + \sum_k \theta_k f_k(z), \qquad P(\text{move} = z) = \frac{e^{U(z)}}{\sum_{z' \text{ empty}} e^{U(z')}}.$$

Here $U(z)$ is the utility of candidate cell z; b_z is a per-cell bias that absorbs position priors holding regardless of context, since corners, star points, and edges are played at different base rates; θ_k are fitted weights; and $f_k(z)$ are geometric features evaluated at the candidate cell. The softmax normalizes the utilities into a distribution over all empty cells, so a higher utility means a more likely move. The features are deliberately simple and entirely local:

- **Distance to the opponent's last stone.** The Chebyshev distance from z to black's most recent move, i.e. how close the candidate sits to where the opponent just played. Its weight is the largest in magnitude and negative, so nearer cells score higher and the model learns to *answer locally*. This single feature is the coupling, written as a coefficient.
- **Distance to my own move two earlier.** The Chebyshev distance from z to white's own previous stone, capturing continuity with one's own play, such as extending or reinforcing one's own shapes. Its weight is also negative but smaller.
- **Local density contrasts.** Within a 5×5 window around z, the difference between the number of one's own stones and the opponent's, including the same contrast measured relative to the opponent's last stone. These say whether the candidate sits in friendly or contested territory, that is, whether one is reinforcing strength or pushing into the opponent's area.

The leading weight encodes the coupling directly: *play near where the opponent just played*. As a predictor, E1 is weak. It captures about 2.4 of the 8.49 bits a move can carry and then plateaus, while a convolutional network reaches about 4.5.

Prediction, however, is the wrong objective. Coupling does not require a good predictor of either stream; it requires a difference between two predictors. I fit a *self* model of stream Y from its own past and a *combined* model that adds the other stream X's past. The improvement in bits measures how much X leaks about Y's future beyond what Y already knew about itself,

$$\text{gap} = \frac{1}{2}\log_2 \frac{\mathrm{Var}(Y_t \mid Y_{<t})}{\mathrm{Var}(Y_t \mid Y_{<t}, X_{<t})} \quad \text{(bits)}.$$

The same shallow model sits on both sides, so E1's weakness as a predictor does not affect its role as a yardstick. To guard against spurious detection I never report the gap alone. Two independent random walks can appear strongly correlated by accident (Phillips, 1986). I therefore compare the gap to a surrogate that preserves each stream's own statistics while destroying the cross-alignment (Theiler et al., 1992), and treat any gap that does not exceed the surrogate as zero.

2.2 The Granger equivalence

The ordinary Granger test fits the same self and combined regressions and asks whether X's lags jointly improve the prediction of Y, through a block-exclusion F-test,

$$F = \frac{(\mathrm{RSS}_{\mathrm{self}} - \mathrm{RSS}_{\mathrm{comb}})/q}{\mathrm{RSS}_{\mathrm{comb}}/(n - k)},$$

where RSS is the residual sum of squares, q the number of added cross-lag regressors, n the sample size, and k the parameters of the combined model. Under the null of no causality, F follows an $F(q, n - k)$ distribution. The gap is a monotone transform of this statistic: with $\text{gap} = -\frac{1}{2}\log_2(1 - \varrho^2)$ and $F \propto \varrho^2/(1 - \varrho^2)$, both rise with the same partial correlation ϱ^2. In information form the gap is Geweke's (1982) measure of linear feedback, and Barnett, Barrett, and Seth (2009) proved that for jointly Gaussian processes Granger causality equals transfer entropy, $\mathrm{TE}_{X\to Y} = \frac{1}{2}F$. On linear-Gaussian data the gap, the F-test, and transfer entropy therefore coincide.

2.3 Probing the limit

To establish where this equivalence holds, I generated linear systems driven by deliberately non-Gaussian innovations (skew-normal, Student-t, and Laplace) and coupled them either through the conditional mean, where X shifts Y's average, or through the conditional variance, where X modulates Y's spread without touching its average. For each system I compared the linear gap to a residual-dependence test based on distance correlation (Székely et al., 2007), which detects dependence of any order and which I calibrated with a permutation null.

2.4 Baduk and its controls

On baduk I measured the black-to-white coupling under two representations. The scalar lens treats each coordinate axis separately. The board-aware lens uses the full two-dimensional move, so any geometric relationship can be discovered. To separate genuine coupling from artifacts I used three controls: a cross-fitted random forest that removes the full nonlinear conditional mean; a no-coupling surrogate that should read null; and a decisive surrogate that carries baduk's exact mean coupling, is bounded to the same grid, but holds constant variance by construction.

3. Results

3.1 The meter reduces to Granger causality

Generalized, the meter reduces exactly to Granger causality. On synthetic Gaussian data its F is identical to standard software, it converges to the analytically known population value as the sample grows, its false-positive rate is correctly sized, and direction is recovered, with the reverse channel reading null. By the equivalence of Section 2.2, no estimator extracts more coupling than this from linear-Gaussian data; transfer entropy, conditional mutual information, and the predictive gap all coincide. This is the wall. For linear-Gaussian coupling, Granger is the limit.

3.2 What the wall depends on

The equivalence requires Gaussianity, and the synthetic battery shows exactly what that buys. First, the wall is linear in the mean rather than Gaussian as such. When coupling acts through the conditional mean, Granger captures it completely, regardless of how non-Gaussian the noise is. What matters is which moment carries the coupling, not the shape of the noise. Coupling carried in the variance is invisible to a mean-based method and appears only in the residual-dependence test; this is volatility spillover, familiar from econometrics (Engle, 1982). Second, away from Gaussianity the gap is a lower bound, $\frac{1}{2}F \leq TE$. The linear gap captures only the second-order part of the coupling, so whatever lives in higher moments or nonlinear structure sits between $\frac{1}{2}F$ and the true transfer entropy. That margin is where a non-Gaussian system can carry coupling past Granger.

3.3 How far past Granger baduk lives

Baduk is not Gaussian; its moves are bounded integers. Its linear coupling is captured exactly by Granger, at about 0.44 bits across both axes. The scalar lens leaves additional dependence that at first resembled a variance-channel breach, but the controls show otherwise. Removing the full nonlinear mean with a cross-fitted forest does not eliminate it, and the no-coupling surrogate reads null, so it is neither misspecification nor pipeline artifact. The decisive bounded, constant-variance surrogate reproduces only a weak version of the signal, so it is not genuine variance coupling either. The leftover is structure the scalar lens cannot express because it cannot see the board.

The board-aware lens recovers an additional 0.06 bits, and the gain is purely nonlinear; a linear two-dimensional lens with cross terms adds nothing. The recovered coupling is local response. White's reply lands within one point of black 22% of the time, within two points 41%, and within three 56%, and the median white-to-black distance is 2.24 against

9.06 under a shuffled null. Two properties fix its character. It is sharp rather than smooth, since smooth polynomial features recover only 4% of it while the forest recovers the rest. And it is shaped by the board edge, since the offset between white and black depends on where black sits. None of this depends on the coordinate origin, because the gap, the forest, and the distance tests rest on differences and variances that are invariant to translation and rescaling.

This corrects an earlier reading in which baduk appeared to sit flat against the wall with no nonlinear structure. That reading was an artifact of the scalar representation, which is too coarse to see two-dimensional shape. There is a small, real, nonlinear coupling, and it is recognizable baduk: contact and approach play, the relationship E1's distance feature already encodes. It is available only because baduk is not Gaussian, exactly the margin where $\frac{1}{2}\mathrm{F}$ falls short of transfer entropy. The residual does not vanish as the lens improves; it shrinks. The scalar-linear lens gives 0.44 bits, the board-aware lens gives 0.50, and the leftover recedes toward the model-free transfer-entropy ceiling without reaching it. The same decomposition applied to a wild-baboon troop (Strandburg-Peshkin et al., 2015), a continuous system, returns the same verdict: a small certified coupling of a few hundredths of a bit, and no dramatic breach.

4. Discussion

I did not build a new detector. I re-derived the Granger and transfer-entropy family from an unusual door, a noise-like board game, and confirmed on a system with ground truth that it is complete for linear coupling and a lower bound otherwise. A plain Granger test matches the meter at lower cost. The instrument is neither new nor better than the family it re-expresses; its value is the route and the controlled measurement.

That measurement has a clear shape. I do not break the wall; I move it. Each richer lens captures a little more of the same coupling and leaves a little less, approaching the model-free ceiling that Gaussian Granger sets as an equality and that a non-Gaussian system sets as a bound.

The decomposition generalizes to any system whose agents respond to one another's positions. The linear coupling splits into a same-axis term (L1) and a cross-axis term (L2), and only a nonlinear geometric term (L3) lies past the wall. It is tempting to call L2 "beyond Granger," but that repeats the error my scalar meter invited. L1 and L2 together are simply proper multivariate Granger causality, evaluated on the full state rather than one axis at a time, so the wall sits at L1 plus L2. Baduk has essentially no L2, since its cross-axis linear term is zero, plus a little L3. A different system can sit elsewhere. In team sports, where players pursue one another in two dimensions, a defender mirroring an attacker is an L2 relationship, while spacing and formation are L3. The prediction is concrete and falsifiable: soccer should show more L2 and basketball more L3, and the meter separates them by the choice of features under the same surrogate-calibrated zero. I have not tested this, lacking tracking data, but the method has already ported across discrete play and animal movement, and team-sport tracking is the same class. The honest claim is portability: the method travels across coordinate-based coupling, while the specific model and lens do not.

The work began because two colors of stone looked like noise and were not. It ends not at a wall I broke but at one I keep pushing back as the lens sharpens, and at a decomposition that should travel to any system where agents answer one another's positions.

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